Random number generators operate by starting with a **seed**, a number between zero and one. When the random number generator is invoked, the seed is passed through a fixed algorithm, resulting in a new number between zero and one. This new number is reported as the *random number*, and is then internally stored to be used as the seed the next time the random number generator is called. The algorithm that transforms the seed into the new random number is often of the form:

EQUATION 2-10
Common algorithm for generating uniformly distributed random numbers between zero and one. In this method, S is the seed, R is the new random number, and a,b,& c are appropriately chosen constants. In words, the quantity $aS+b$ is divided by c, and the remainder is taken as R.

$$R = (aS + b) \text{ modulo } c$$

In this manner, a continuous sequence of random numbers can be generated, all starting from the same seed. This allows a program to be run multiple times using exactly the same random number sequences. If you want the random number sequence to change, most languages have a provision for **reseeding** the random number generator, allowing you to choose the number first used as the seed. A common technique is to use the *time* (as indicated by the system's clock) as the seed, thus providing a new sequence each time the program is run.

From a pure mathematical view, the numbers generated in this way cannot be absolutely random since each number is fully determined by the *previous* number. The term **pseudo-random** is often used to describe this situation. However, this is not something you should be concerned with. The sequences generated by random number generators are statistically random to an exceedingly high degree. It is very unlikely that you will encounter a situation where they are not adequate.

Precision and Accuracy

Precision and accuracy are terms used to describe systems and methods that *measure*, *estimate*, or *predict*. In all these cases, there is some parameter you wish to know the value of. This is called the **true value**, or simply, **truth**. The method provides a **measured value**, that you want to be as close to the true value as possible. *Precision* and *accuracy* are ways of describing the error that can exist between these two values.

Unfortunately, precision and accuracy are used interchangeably in non-technical settings. In fact, dictionaries define them by referring to each other! In spite of this, science and engineering have very specific definitions for each. You should make a point of using the terms correctly, and quietly tolerate others when they used them incorrectly.